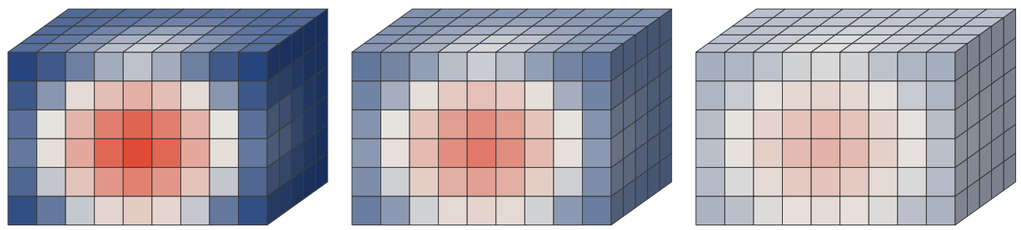
1 Block value changes over time

Period 1 Period 2 Period t



Early periods → Late periods

$v_{j,t}$

Low value — High value

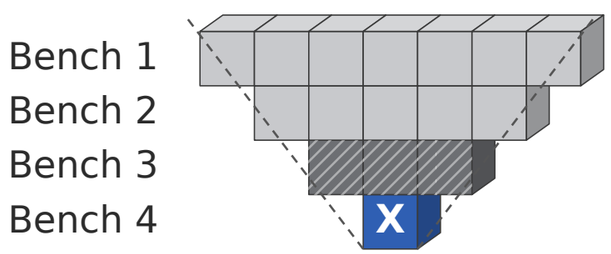
0

$$v_{j,t} = v_j/(1+d)^{t-1}$$

2 Precedence constraint

Slope stability, 2D section

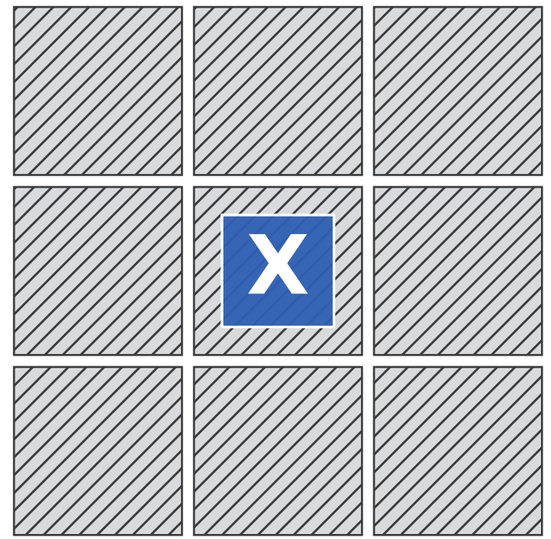
- Target j
- Immediate Predecessor
- Allowable extraction cone



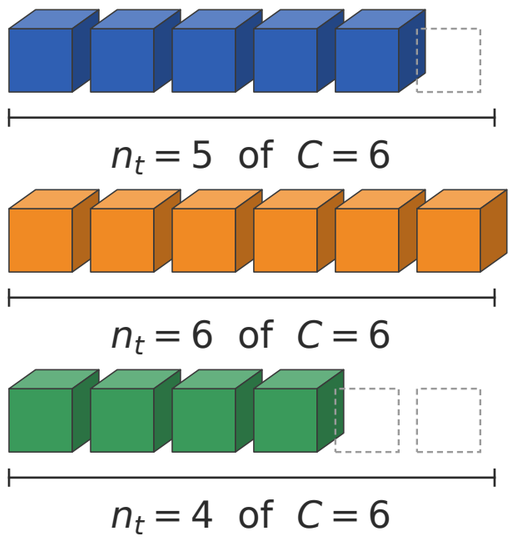
Cone extracted before j

3 Top view

Immediate Predecessors



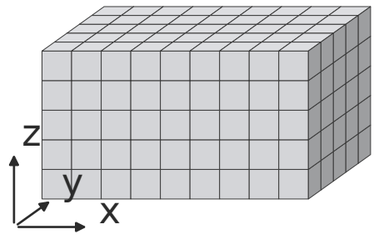
4 Mining capacity constraint



Dashed: unused capacity (slack)

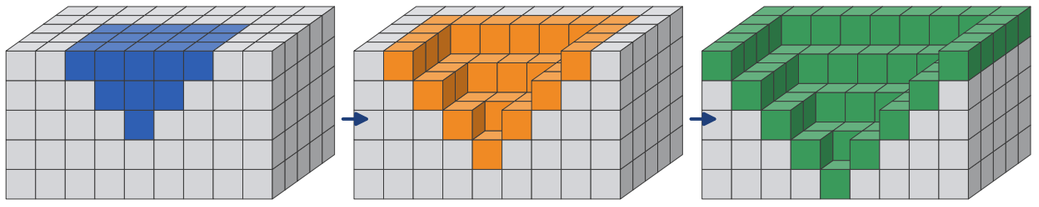
At most C blocks per period: $n_t \leq C$

5 Discretization and evolution over time



Block model (spatial)

Period 1 Period 2 Period 3



Colored: extracted in this period
Earlier periods already removed